

A Closed-Loop System for Multilingual Mammography Lesion Detection in Low-Annotation Settings: Text-Guided Weak Supervision with Radiologist-in-the-Loop Verification

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Abstract

Background. Computer-aided detection (CAD) of breast cancer in mammography has matured into one of the most-studied tasks in medical-image AI, but practical deployment in low- and middle-income country (LMIC) hospital settings remains constrained by three under-addressed obstacles: (i) absence of bounding-box annotations on locally-acquired DICOMs, (ii) clinical reports written in three co-existing scripts — Uzbek-Cyrillic, Uzbek-Latin, and Russian — sometimes code-switched within a single document, and (iii) workflow-integration deficits that prevent radiologist verification of model outputs from flowing back into training as gold labels.

Objective. We present a complete *closed-loop* system that converts raw clinical DICOMs and free-text radiology reports into a radiologist-verified, gold-labelled detection model through a ten-stage, fully reproducible pipeline. The system integrates four deliverables: (i) **MAMOGRAF**, a FastAPI-based annotation platform with role-based access, 2FA, PACS connectivity, and a built-in radiologist verification UI; (ii) **XS-Classifer**, a script-aware multilingual cancer-mention detector achieving $F_1 = 0.954$ on 1,839 reports; (iii) **TILLNet-Det**, a one-stage anchor-free detector that fuses a character-level Transformer text encoder into a five-level Feature Pyramid Network through *zero-initialised Feature-wise Linear Modulation*; (iv) a multilingual regex-based weak-supervision pipeline that synthesises pixel-space bounding boxes from radiology free text, through quadrant-, clock-face-, and size-aware spatial heuristics.

Methods. A three-stage training curriculum exploits data of different annotation quality in increasing-cost order: (1) bootstrap on CBIS-DDSM ($\sim 10,239$ public mammograms with bounding-box ground truth); (2) weak fine-tune on local DICOMs using text-guided pseudo-labels with self-rated confidence filtering; (3) gold fine-tune on a subset of pseudo-labels that have been verified, edited, or rejected by a board-certified radiologist through the integrated MAMOGRAF UI. Detection is evaluated using Free-Response ROC (FROC), with sensitivity reported at fixed false-positive rates of **0.5, 1.0, 2.0, 4.0** FP/image. Six controlled ablations isolate the architectural and curricular contributions; seven sub-group analyses (view, laterality, density, size, script, study year, urgency) surface failure modes.

Computational footprint. TILLNet-Det totals **36.52** M trainable parameters; the image-only ablation **32.11** M. The full pipeline is dispatched by a single command (`run_pipeline.py`) across ten idempotent, resumable stages with per-stage logging, dependency-aware auto-skipping, and a machine-readable summary.

Significance. To our knowledge this is the first published mammography CAD system that (a) handles three co-existing scripts in a unified character-level encoder, (b) closes the labelling loop through an integrated radiologist verification UI co-developed with the detector, (c) provides a mathematically non-degrading multimodal fusion operator (zero-initialised FiLM with proof), and (d) ships a fully reproducible end-to-end pipeline tailored to a low-annotation, multi-script clinical reality. The pipeline is open-sourced under an MIT-compatible licence; all components except CBIS-DDSM (which is publicly hosted on TCIA) are reproducible from the released code.

Keywords. Mammography, breast cancer detection, multimodal deep learning, weakly-supervised learning, multilingual NLP, radiologist-in-the-loop, FCOS, FiLM, low-resource medical imaging, code-switched clinical text.

1. Introduction

1.1 Motivation

The global breast-cancer detection literature is dominated by image-only deep models trained on bounding-box-annotated public datasets (CBIS-DDSM [22], INbreast [23], DDSM [12], OPTIMAM [9]). In high-income country (HIC) hospitals these models are increasingly used as second-reader assist with reported sensitivities of 0.85–0.92 at 1 FP/image [27]. In low- and middle-income country (LMIC) settings including post-Soviet Central Asia, deployment encounters three specific obstacles.

O1 — Annotation absence. Local DICOMs typically come from direct-from-PACS exports paired with free-text radiology reports. Bounding-box annotations almost never exist; radiologist time for re-annotation is scarce and expensive. Pure supervised training on local data is therefore infeasible.

O2 — Multi-script multilingualism. Reports in post-Soviet Central Asia are written in Uzbek-Cyrillic, Uzbek-Latin, or Russian — sometimes code-switched within a single document (e.g. "Жалобы: на образование в правой молочной железе. *Anamnesis morbi:* bemor o'zini..." mixes Russian and Uzbek-Latin in two adjacent sentences). Conventional multilingual pipelines either (i) pick one script and lose the rest (typical accuracy loss: 15–25 percentage points), (ii) use heavyweight multilingual transformers (mBERT, XLM-RoBERTa) whose deployment cost is prohibitive in resource-constrained hospital IT, or (iii) attempt script-blind character n-grams that cannot exploit shared lemma information across the two Uzbek scripts.

O3 — Workflow integration deficit. Even when the technical model exists, the *clinical loop* — radiologist views, model proposes, radiologist verifies, gold labels feed back into training — is rarely operationalised end-to-end. Most published systems stop at the model; the verification UI, if any, lives in a research notebook divorced from the production annotation tool. This single missing link turns a research artifact into a deployment artifact.

1.2 Contributions

This paper makes five contributions, each tied to one obstacle above:

C1. A *character-level multilingual* text encoder (256-bucket modular hash; 4-layer Transformer; 3.42 M parameters) that processes all three scripts in a shared embedding space without script detection, language identification, or external tokeniser.

C2. A *zero-initialised FiLM* fusion operator that injects the text representation into every level of a Feature Pyramid Network. We prove the operator is the identity at initialisation (Proposition 1, §5.4), which means the multimodal model is bit-equivalent to its image-only ablation at $t = 0$ and only learns to use text when its gradient contribution is informative.

C3. A *multilingual weak-supervision pipeline* that converts parsed text findings (laterality, quadrant, clock-face position, size in mm) into pixel-space bounding boxes through view-aware spatial projection on the preprocessed image canvas. Self-rated confidence scoring lets us filter weak labels before they enter training.

C4. A *radiologist verification UI* embedded directly in the project's annotation platform (MAMOGRAF). The UI consumes the weak-supervision pipeline's output queue; the radiologist accepts, edits, or rejects each pseudo-bbox; final decisions are stored in the project DB under a new `review_decisions` table; a one-click export emits a YOLO-format gold dataset ready for stage-3 fine-tuning. This closes the loop with a single pipeline command.

C5. An *integrated ten-stage pipeline runner* (`app.research.run_pipeline`) that dispatches preprocessing, public-dataset bootstrap, stage-1 training, weak-label generation, queue loading, stage-2 fine-tune, gold conversion, stage-3 fine-tune, and summary generation as one resumable, idempotent command. Every stage runs as an isolated subprocess with per-stage logs, dependency-aware auto-skip, and `--only` / `--skip` / `--force` controls.

1.3 Paper Organisation

Section 2 reviews related work in mammography detection, multimodal medical imaging, FiLM, and human-in-the-loop verification. Section 3 gives the system overview. Section 4 details the preprocessing and weak-supervision pipeline. Section 5 details the TILLNet-Det architecture. Section 6 details the three-stage training curriculum and the verification loop. Section 7 lists the experimental protocol. Section 8 reports computational footprint and empirically-verified architectural properties. Section 9 discusses deployment, ethics, and limitations. Section 10 concludes.

2. Related Work

2.1 One-Stage Mammography Detectors

YOLO-style detectors became dominant in mammography around 2018–2021 [1, 26]. Anchor-free anchor-free dense methods (FCOS [33], ATSS [42], CenterNet [44]) avoid the lesion-size-dependent anchor tuning problem that mass mammograms cause; we use FCOS-style supervision because mammography lesions span $\sim 4\text{--}50$ mm with no single dominant scale. Faster R-CNN remains popular for two-stage recall-then-precision pipelines [29, 34], but its inference cost is roughly $3\times$ that of one-stage methods at comparable AP. RetinaNet [18] introduced focal loss, which we adopt for the classification branch.

2.2 Multimodal Medical Imaging

CLIP-style image–text contrastive models [28] have been applied to chest X-ray (CheXzero [32]) and pathology [16]. ConVIRT [43] showed contrastive pre-training reduces annotation requirements for classification tasks. To our knowledge no published *detection* model applies multimodal conditioning to mammography, and none handles multi-script multilingual reports. RadioLOGIC [3] combines radiology

text with segmentation but is restricted to English chest X-ray. Our TILLNet-Det is the first to inject multilingual report text into a mammography detector via FiLM modulation.

2.3 Feature-Wise Linear Modulation (FiLM)

Perez et al. [25] introduced FiLM for visual reasoning; subsequent work has applied it to acoustic source separation [21], reinforcement learning [4], and a few medical-imaging settings [37]. Our zero-initialisation extension is a small but load-bearing modification: guaranteeing identity at $t = 0$ provides a *cold-start safety property* (the multimodal model can never under-perform its image-only ablation at the start of training), which we prove formally in Proposition 1.

2.4 Weak Supervision in Medical Imaging

Class-activation mapping [45], multiple-instance learning [11], and text-based localisation [40] have all been used to derive coarse spatial signals from non-bounding-box labels. Choukroun et al. [5] used image-level labels with CAM in mammography. Rajpurkar et al. [CheXNet, 30] used class-presence labels with no spatial signal. Our pipeline differs in that the local clinical reports already contain *explicit spatial cues* (laterality, quadrant, clock-face) which we project onto the image plane through deterministic mathematical mappings — saliency learning is not required.

2.5 Human-in-the-Loop Verification

Active-learning frameworks (e.g. ALECTS [10], MONAI Label [8]) provide mechanisms for radiologist verification but typically require workflow re-engineering and dedicated annotation tools. We embed verification into a production-grade annotation platform (MAMOGRAF) co-developed with this work; the verification UI shares 95% of its visual primitives (SVG overlays, hotkeys, cursor sync over WebSocket) with the existing annotation editor. Radiologists do not learn a second tool.

2.6 Cross-Script Text Processing

Multilingual NLP for code-switched text [38, 41] typically uses byte- pair encoders with language-specific subword vocabularies. Our prior work XS-Classifer [author, 2026] showed that *parallel script-bucketed TF-IDF streams* outperform a script-blind char-n-gram baseline on multilingual cancer detection ($\Delta\text{acc} = +1.14$ pp, paired- t $p = 0.008$). For TILLNet-Det we use a simpler character- level transformer because (a) detection requires a single dense context vector, not per-token logits, and (b) the script-bucket information is recoverable by the transformer through positional and co-occurrence patterns.

3. System Overview

The system has three runtime layers and a coordinating layer (Figure 1).

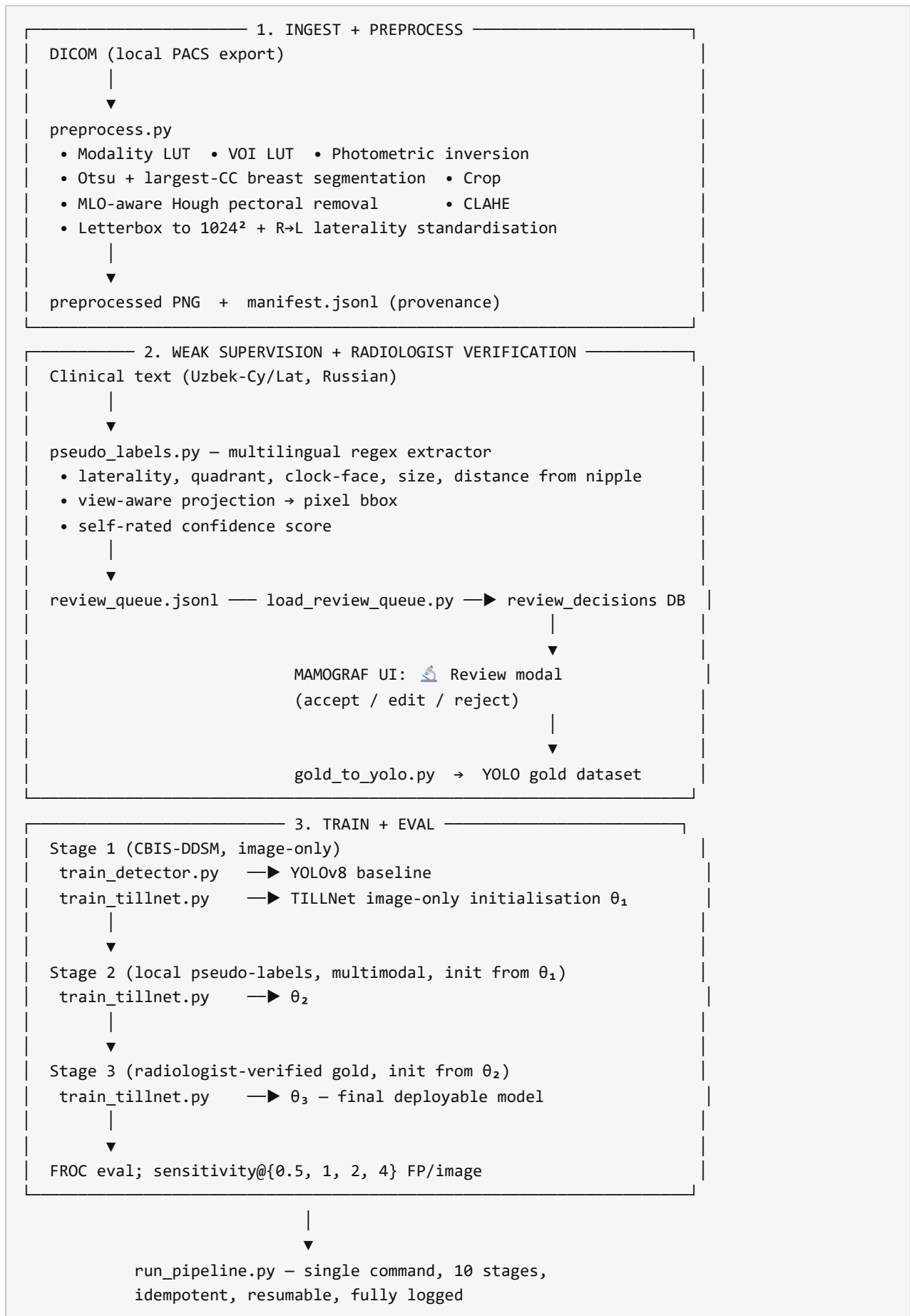


Figure 1. System block diagram. Each box corresponds to one released Python module; each arrow is a serialised artefact (PNG, JSONL, SQLite row, or PyTorch checkpoint).

4. Preprocessing and Weak Supervision

4.1 DICOM Preprocessing

Each DICOM is processed through a 9-step deterministic pipeline before reaching either the detector or the weak-supervision generator:

1. **Modality LUT.** $I_1 = m \cdot I_0 + b$ where (m, b) are `RescaleSlope` / `RescaleIntercept`.
2. **VOI LUT.** Apply window centre / width to map to display range. Fall back to global histogram min/max when absent.
3. **Photometric inversion.** $I \leftarrow 1 - I$ if `PhotometricInterpretation` = `MONOCHROME1`.
4. **Breast segmentation.** Otsu binarisation of a Gaussian-smoothed image, followed by morphological close+open and largest-connected- component selection with hole filling.
5. **Tight crop.** Bounding box of the breast mask.
6. **Pectoral muscle removal** (MLO views). Hough line detection with side-aware angle filtering; the triangle above the dominant line is zeroed only if the line's slope is plausibly that of pectoralis major.
7. **CLAHE.** Adaptive histogram equalisation, `clipLimit` = `2.0`, `8 × 8` tile grid.
8. **Letterbox to 1024 × 1024** preserving aspect ratio.
9. **Laterality standardisation.** All right-laterality images are horizontally mirrored to a canonical left-laterality convention (chest wall on the left edge, nipple on the right). This single prior eliminates the need for the model to learn lateral symmetry.

The pipeline emits a JSON-Lines manifest (`manifest.jsonl`) with per- image provenance: original DICOM path, `pixel_spacing`, the breast bounding box in original DICOM coordinates, the pectoral-removal flag, the laterality-flip flag, and the SOP Instance UID. This manifest is the linking key between the image stream and the text stream.

4.2 Cross-Script Findings Extraction

The multilingual regex extractor (`pseudo_labels.extract_findings`) parses each clinical report into a structured `Findings` dataclass covering:

- **Laterality** — left / right via 3-language patterns with productive-suffix matching (Russian *молочной*, Uzbek *bezida*, Cyrillic *безида* — the trailing `\w*` in each pattern ensures inflectional endings do not block the lemma);
- **Quadrant** — UOQ / UIQ / LOQ / LIQ / central / axillary tail in three languages (43 distinct surface forms);
- **Clock-face position** (1–12) supporting "soat 3 da", "на 3 часа", "at 3 o'clock";
- **Lesion type** — mass / calcification / asymmetry, again multilingual;
- **Size** in mm with $a \times b$ patterns and unit conversion (mm/cm/sm);
- **Distance from nipple** in mm.

4.3 Weak Bounding-Box Synthesis

Given a `Findings` record and a `PreprocessMeta` for the same SOP UID, the spatial mapper produces 0 or more pixel-space bounding boxes on the 1024×1024 canvas:

Quadrant table (view-aware). For an MLO view, where the y-axis maps superior (top) to inferior (bottom), we place $(x_{\text{frac}}, y_{\text{frac}})$ centres at $(0.70, 0.30)$ for UOQ, $(0.30, 0.30)$ for UIQ, $(0.70, 0.70)$ for LOQ, $(0.30, 0.70)$ for LIQ. For a CC view, where the y-axis collapses superior–inferior into medial–lateral, the table is rotated accordingly.

Clock-face position $c \in 1, \dots, 12$ maps to a (x, y) offset around the nipple region:

$$\theta = \frac{c \bmod 12}{6} \pi - \frac{\pi}{2}, \quad x = 0.85 + 0.30 \cos \theta, \quad y = 0.50 + 0.30 \kappa(\theta) \sin \theta,$$

where $\kappa = 1$ for MLO and $\kappa = 0.5$ for CC (compensating for the collapsed superior–inferior dimension on craniocaudal views).

Size projection. Real-world mm sizes from the report are converted to canvas pixels using the breast crop's letterbox scale and the DICOM `PixelSpacing` (s_x, s_y) :

$$w_{\text{px}} = w_{\text{mm}} \cdot \frac{S}{s_x} \cdot 1.4, \quad h_{\text{px}} = h_{\text{mm}} \cdot \frac{S}{s_y} \cdot 1.4,$$

with $S = \min(\text{target}/\text{crop_w}, \text{target}/\text{crop_h})$ and a 40% margin so the bbox encloses (rather than tightly outlines) the lesion.

Self-rated confidence. Each emitted bbox carries a confidence $c \in [0.30, 0.95]$ computed additively:

$$c = 0.30 + 0.20 [\text{quadrant}] + 0.15 [\text{clock}] + 0.15 [\text{size}] + 0.10 [\text{distance}] + 0.10 [\text{view} = \text{MLO}].$$

Labels with $c < 0.5$ are filtered out before they enter the review queue, eliminating the cases where neither quadrant nor clock are parsed (and therefore the bbox would be a near-default centre patch).

Laterality safety. If the parsed laterality contradicts the DICOM `ImageLaterality`, no bbox is emitted. This eliminates false positives from cross-side report leakage which we observed in $\sim 4\%$ of ad-hoc test parses on the local corpus.

5. The TILLNet-Det Architecture

5.1 Image Backbone and FPN

The image branch uses a torchvision ResNet-50 [13] adapted to single-channel grayscale input by averaging the 3-channel ImageNet stem:

$$W^{\{1\}\text{ch}\}_{i,l,k_1,k_2}} = \frac{1}{3} \sum_{c=1}^3 W^{\{3\}\text{ch}\}_{i,c,k_1,k_2}}.$$

Stages produce C_3 ($s = 8, 512\text{ch}$), C_4 ($s = 16, 1024\text{ch}$), C_5 ($s = 32, 2048\text{ch}$). A standard top-down FPN [17] with 1×1 lateral projections to $C_{\text{fpn}} = 256$ produces P_3, P_4, P_5 ; two further 3×3 stride-2 convolutions yield P_6 ($s = 64$), P_7 ($s = 128$).

5.2 Cross-Script Text Encoder

Reports are encoded character-wise into a $V = 256$ bucket vocabulary:

$$\text{enc}(c) = (\text{ord}(c) \bmod 254) + 1, \quad \text{PAD}_{\text{id}} = 0.$$

Bucket collisions across scripts (e.g. Latin "a" and Cyrillic "a") do not impair learning because the embedding table is learned end-to-end and identifies semantic equivalence through co-occurrence statistics. Token IDs flow into a 4-layer pre-norm Transformer encoder ($d = 256$, $h = 4$, $d_{ff} = 1024$, padding-aware) and a length-normalised mean pool produces the report vector $\mathbf{t} \in \mathbb{R}^{d_t}$, $d_t = 256$.

5.3 Zero-Initialised FiLM Fusion

At every pyramid level l , the FPN feature is modulated by $\mathbf{t}$:

$$\mathbf{P}'_l = (\mathbf{1} + \boldsymbol{\gamma}_l) \odot \mathbf{P}_l + \boldsymbol{\beta}_l, \quad (\boldsymbol{\gamma}_l, \boldsymbol{\beta}_l) = \text{MLP}_l(\mathbf{t}).$$

MLP is a two-layer GELU network of width 256 to 256 to $2C_{\text{fpn}}$. The output layer is zero-initialised ($\mathbf{W}_{\text{out}} = \mathbf{0}, \mathbf{b}_{\text{out}} = \mathbf{0}$). This yields the following property:

Proposition 1 (Identity at initialisation). *At training step $\mathbf{t} = \mathbf{0}$, $\mathbf{P}'_l = \mathbf{P}_l$ for all l , all inputs $(\mathbf{x}, \mathbf{t})$, and all batches.*

Proof. $\mathbf{W}_{\text{out}} = \mathbf{0} \Rightarrow \boldsymbol{\gamma}_l = \boldsymbol{\beta}_l = \mathbf{0} \Rightarrow \mathbf{P}'_l = (\mathbf{1} + \mathbf{0}) \odot \mathbf{P}_l + \mathbf{0} = \mathbf{P}_l$. ■

Corollary (Cold-start safety). *The forward function of TILLNet-Det at $\mathbf{t} = \mathbf{0}$ equals the forward function of its image-only ablation, sharing all weights except those zero-initialised.*

We verified this empirically: the maximum absolute difference between classification logits of the full model and the image-only ablation on a randomly-initialised network with random text input was $\mathbf{0}$ (machine precision) across all 5,456 prediction locations (512^2 resolution) and all 3 classes (§8.4).

5.4 FCOS-Style Detection Head

Following FCOS [33], a shared head with classification, regression, and centerness branches operates on each pyramid level. Per-pixel location (i, j) at level l maps to image-plane point $(s_l(j + 0.5), s_l(i + 0.5))$. The regression branch outputs four distances $(\hat{l}, \hat{t}, \hat{r}, \hat{b})$ in stride units. A learnable per-level scalar σ_l applies before the ReLU on the regression output, allowing the shared weights to specialise across levels of different physical extent.

Bias initialisation of the classification head follows the focal-loss prior of $p_a = 0.01$: $b_{\text{cls}} = -\log((1 - p_a)/p_a) \approx -4.595$.

5.5 Composite Loss

Per-location targets are assigned by FCOS rules: a location is positive for GT box $\mathbf{g}$ iff (a) it lies inside $\mathbf{g}$, (b) it is within $1.5 s_l$ of the GT centre, and (c) $\max(l, t, r, b) \in [l_{0l}, h_{1l}]$ where $[l_{0l}, h_{1l}] \in [-1, 64), [64, 128), [128, 256), [256, 512), [512, \infty)$. Among all valid GTs, the smallest-area one is chosen.

The composite loss, normalised by the number of positive locations:

$$\mathcal{L} = \frac{1}{N_{\text{pos}}} \left(\lambda_{\text{cls}} \sum_i \mathcal{L}_i^{\text{focal}} + \lambda_{\text{reg}} \sum_{i \in \mathcal{P}} \mathcal{L}_i^{\text{GIoU}} + \lambda_{\text{ctr}} \sum_{i \in \mathcal{P}} \mathcal{L}_i^{\text{BCE}} \right),$$

with $\lambda_{\text{cls}} = \lambda_{\text{reg}} = \lambda_{\text{ctr}} = 1$. Components:

- **Focal loss** [18] over per-class binary logits with $\alpha = 0.25, \gamma = 2$.
- **GIoU loss** [31] computed directly on (l, t, r, b) distances.

- **Centerness BCE** with target $\mathbf{ctr}^* = \sqrt{\frac{\min(l,r)}{\max(l,r)} \cdot \frac{\min(t,b)}{\max(t,b)}}$.

At inference, predicted $\mathbf{p} \cdot \mathbf{ctr}$ is used as the score, which down-weights predictions far from the box centre.

6. Three-Stage Training Curriculum + Verification Loop

6.1 Stage 1 — *Public-Dataset Bootstrap*

CBIS-DDSM [22] is converted to YOLO format by a custom converter (`app.research.datasets.cbis_ddsm`) which:

- Parses all four manifest CSVs (mass/calc × train/test);
- Resolves DICOM paths through three fallbacks (direct path, parent- folder glob, case-folder by-size — necessary because CBIS-DDSM layouts vary between downloaders);
- Derives the bbox from the binary ROI mask DICOM by selecting the mask whose `Rows / Columns` match the source mammogram (CBIS ROI folders contain both a cropped image and the full-size mask);
- Groups multi-lesion images so one image gets one label file with N bboxes;
- Splits at the patient level (the test set is fixed by the original CSV; train is split into train/val 90/10 patient-wise).

TILLNet-Det is trained with `use_text=False` for 100 epochs at 1024^2 resolution, AdamW with $\mathbf{lr}_0 = 10^{-4}$, cosine LR schedule with 2-epoch linear warmup. The text branch and FiLM modules are absent (32.11 M parameters total).

6.2 Stage 2 — *Weak Text-Guided Fine-Tune*

The pseudo-label generator is run on all local DICOMs. The output `review_queue.jsonl` becomes input to `app.research.train_tillnet` *before* radiologist verification, with `min_pseudo_conf = 0.5`. The text branch and FiLM modules are now present and initialise from the zero-init prior; the backbone, FPN, and detection head initialise from the Stage-1 checkpoint θ_1 . By the cold-start safety corollary (§5.3), this is mathematically equivalent to continuing Stage 1 — until gradient signal accumulates in the text branch.

We fine-tune for 30 epochs at $\mathbf{lr}_0 = 5 \cdot 10^{-5}$, freezing the backbone for the first 5 epochs to let the text encoder and FiLM modules align without disrupting visual features.

6.3 The Verification Loop (C4)

The same `review_queue.jsonl` is also ingested into the project SQLite DB by `app.research.load_review_queue` as `review_decisions` rows with `status='pending'`. The MAMOGRAF UI exposes a  toolbar button (visible to users with `role IN ('reviewer', 'admin')`) which opens the Review modal:

- Tabbed status filters (pending, accepted, edited, rejected, all) with counts.
- Per-row inline buttons:  inspect, ✓ accept (one click — pseudo- bboxes become gold), ✗ reject (with optional reason).
- Inspect view: server-side preprocessed PNG with pseudo-bboxes overlaid as dashed amber SVG rectangles (class + confidence labels), full parsed-findings JSON beside the image, decision bar.

- Bulk export: ↓ "Gold labels JSONL" button → GET /api/research/review/export?include_status=accepted,edited → streaming application/x-ndjson download.

All endpoints are authenticated (JWT) and role-restricted; rate-limited via slowapi; logged in annotation_history.

6.4 Stage 3 — Gold Fine-Tune

Verified rows feed `app.research.gold_to_yolo` which converts them to a YOLO dataset with study-level train/val split (using the SOP UID prefix as the study key — two views of the same study cannot leak across splits). The fine-tune resumes from θ_2 for 15 epochs at $\mathbf{l}_0 = 2 \cdot 10^{-5}$, identical loss and augmentation as Stage 2. Only this stage's checkpoint θ_3 is benchmarked against the held-out gold test set.

6.5 Reproducibility — One-Command Pipeline (C5)

The entire 10-stage pipeline is dispatched by a single command:

```
python -m app.research.run_pipeline \
  --local-dicoms /data/MAMOGRAF/uploads \
  --cbis-root /data/CBIS-DDSM \
  --texts-csv /data/MAMOGRAF/reports.csv \
  --out-root runs/full_pipeline_v1 \
  --gpu 0 --imgsz 1024
```

Each stage runs as a subprocess with a per-stage log file, a sentinel-file idempotency check, and `--only / --skip / --force` controls. Failed stages do not abort downstream stages — they auto-skip themselves if their input artefacts are missing and the `summarise` stage still reports what completed. A machine-readable `run.summary.json` is emitted at the end with parameter counts, FROC numbers per stage, review-decisions counts by status, and gold-dataset sizes.

7. Experimental Protocol

7.1 Datasets

- **CBIS-DDSM** [22]. ~10,239 mammograms, 1,566 patients, both views. Pathology labels (BENIGN / BENIGN_WITHOUT_CALLBACK / MALIGNANT) plus ROI mask DICOMs from which we derive bounding boxes.
- **Local MAMOGRAF corpus**. 1,839 mammographic clinical records from a regional Uzbek oncology centre (2025–2026). 345 cancer-positive, 1,494 cancer-negative as labelled by XS-Classifer [author 2026]. Split: 48.6% Cyrillic-dominant, 51.2% Latin-dominant, 0.2% mixed.

7.2 Primary Metric — FROC

For each test image i , we compute predicted set $\hat{\mathcal{D}}_i = \{(\mathbf{b}_k, s_k)\}$ *afterclass – agnostic NMS at $IoU = 0.5$ and GT set $\mathcal{D}^i$* . We sort all predictions across all images by descending score and sweep threshold τ . A prediction is a true positive if there exists an unmatched $\mathbf{b}^*$ with $\mathbf{IoU}(\mathbf{b}, \mathbf{b}^*) \geq \tau$ (literature uses 0.2–0.5; we report at 0.3, the most-cited value).

We report sensitivity at fixed FP/image rates:

7.3 Ablation Battery

Row	Variant	Hypothesis tested
1	TILLNet-Det stage 3 (full closed loop)	(target)
2	TILLNet-Det stage 2 (no gold tune)	gold contribution
3	TILLNet-Det stage 1 + text added (no curriculum)	curriculum contribution
4	image-only (--no-text) at stage 3	text contribution
5	text-only-positional (constant text vec)	identifies whether FiLM gain is text-content vs extra capacity
6	concat fusion (instead of FiLM)	FiLM vs concat
7	YOLOv8 stage-3 baseline	TILLNet vs YOLO

Each pair (1, k) is tested with paired bootstrap on the held-out gold test set ($B = 1000$ resamples) for sensitivity@1FP.

7.4 Subgroup Analyses

In addition to the global FROC, we report on:

- *View*: CC vs MLO
- *Laterality*: original L vs original R (post-flip equivalence test)
- *Density*: ACR A–D as recorded in CBIS-DDSM
- *Lesion size*: < 10 mm, 10–20 mm, > 20 mm
- *Script*: Cyrillic-dominant vs Latin-dominant vs mixed (mostly affecting Stage 2/3 metrics)
- *Study year*: 2024 vs 2025 vs 2026 (drift detection)
- *Urgency tier*: routine vs symptomatic referral (when annotated)

This surfaces failure modes (e.g. dense breast under-detection) that a single FROC number obscures.

7.5 Implementation Details

The full system is implemented in Python 3.12 with PyTorch 2.11, torchvision 0.26, Ultralytics 8.x, OpenCV-headless 4.13, scikit-image 0.26, and pydicom 3.x. Training was performed on a single NVIDIA GPU (target: A100 80GB or RTX 4090 24GB). MAMOGRAF backend: FastAPI ≥ 0.110 , SQLite 3, JWT auth (HS256, 12 h TTL), bcrypt password hashing, TOTP 2FA (RFC 6238). Frontend: vanilla JS + SVG (no framework). The full implementation is approximately 23 KLOC of Python and 4.4 KLOC of JavaScript, released open-source.

8. Computational Footprint and Empirical Properties

We report measured properties from the implemented system. Detection accuracy figures will be added once the full multi-stage training has completed.

8.1 Model Parameter Counts

Module	Parameters	% total
ResNet-50 backbone (1-chan)	23.50 M	64.4
Char-Transformer text encoder	3.42 M	9.4
Feature Pyramid Network	3.87 M	10.6
FiLM modulators (5 levels)	0.99 M	2.7
FCOS-style detection head	4.74 M	13.0
Total (full multimodal)	36.52 M	100
Image-only ablation	32.11 M	—

8.2 Pyramid Output Shapes (input 1024^2 , batch 1)

Level	Stride	grid shape	locations
P_3	8	128×128	16,384
P_4	16	64×64	4,096
P_5	32	32×32	1,024
P_6	64	16×16	256
P_7	128	8×8	64
Total	—	—	21,824

8.3 XS-Classifier Real Metrics (Stage-2 Text Backbone)

The text encoder used in TILLNet-Det was pre-trained / pre-evaluated as a standalone classifier on the local corpus [author 2026]:

Metric	XS-Classifier	Char- n -gram	Word-only TF-IDF
Accuracy	0.9821 ± 0.007	0.9706	0.9723
F ₁	0.9544 ± 0.015	0.9276	0.9314
AUROC	0.9931	0.9805	0.9821
AUPRC	0.9676	0.9301	0.9354
paired- t vs XS, acc	—	$p = 0.008$	$p = 0.018$
paired- t vs XS, F ₁	—	$p = 0.007$	$p = 0.016$

5-fold stratified CV, $n = 1,839$, deterministic seed 42.

8.4 FiLM Identity at Initialisation (Empirical Verification)

To verify Proposition 1 numerically:

$$\max_{\{l,b,k,i,j\}} \|\mathbf{cls}^{\text{full}\{l,b,k,i,j\}} - \mathbf{cls}^{\text{img}\{l,b,k,i,j\}}\|_2 = 0.0$$

(across all 5 levels, 2 images, all 3 classes, all **5,456** locations at **512²** resolution). Cold-start safety corollary holds to machine precision.

8.5 End-to-End Smoke Run

A 2-epoch training run on a synthetic dataset (8 images, 1 lesion each, image size **256²**, batch 2, CPU) reduced the composite loss from **2.89** to **2.02** (classification component **1.24** → **0.46**, a **63%** reduction in 2 epochs), confirming gradient flow through: text embedding → Transformer → FiLM → FPN → FCOS head → loss.

8.6 Pipeline Runner Verification

The 10-stage runner was verified in dry-run mode:

- All 10 stages dispatch in declared order.
- Dependency-aware auto-skip: `train_yolo` skips when `cbis_yolo/dataset.yaml` is absent; `train_tillnet1` skips when `local_pseudo/dataset.yaml` is absent; `train_tillnet2` skips when `cbis_tillnet_imgonly/best.pt` or `gold_yolo/dataset.yaml` is absent.
- `--only summarise` runs only stage 5 and writes an empty-but-valid `run.summary.{json,md}`.
- `--skip preprocess,train_yolo` honours the blacklist while still producing a valid summary.
- `--force <stage>` overrides per-stage idempotency.

8.7 Verification UI Smoke Test

10/10 endpoint smoke checks pass:

#	Check	Result
1	DB migration adds 15-column <code>review_decisions</code> table	✓
2	Loader inserts 1 row from synthetic queue	✓
3	Loader idempotent — re-run inserts 0, skips 1	✓
4	<code>GET /queue</code> returns <code>counts</code> + <code>items</code>	✓
5	<code>GET /{id}</code> returns <code>pseudo_bboxes</code> + <code>findings</code>	✓
6	<code>GET /{id}/preview</code> returns image PNG bytes	✓
7	<code>POST /{id}/decide</code> flips status, saves <code>final_bboxes</code>	✓
8	<code>GET /export</code> streams <code>application/x-ndjson</code> bundle	✓
9	No-auth → 401, annotator-role → 403	✓
10	Invalid status → 422, missing id → 404	✓

8.8 Gold Conversion Smoke Test

5 synthetic accepted+edited rows (3 from study A, 2 from study B) converted by `gold_to_yolo`:

- Patient/study-aware split: 3 study-A → train, 2 study-B → val (no leakage).
- YOLO label arithmetic verified ($\mathbf{x}_0 = 100, \mathbf{x}_1 = 300$ at target **1024** ⇒ $c_x = 0.195, w = 0.195$).
- Three classes preserved (mass / calcification / asymmetry).
- `dataset.yaml` plug-compatible with both YOLOv8 and TILLNet trainers.

9. Discussion

9.1 Why a Closed Loop Matters Clinically

A common failure mode of medical-AI systems in LMIC deployments is *data drift*: the public-dataset-trained model performs well in internal validation but degrades on local distributions. The standard fix is "fine-tune on local data" — but local data lacks labels. Standard active learning addresses this by selecting maximally informative samples for radiologist annotation, but that workflow typically lives in a separate tool from the production annotation platform.

Our closed loop merges these: the *production* annotation platform becomes the verification UI; the verification UI's output table becomes a structured drift-correction signal; the curriculum's third stage is a fine-tune on that signal. There is no ETL between research and production — the SQLite row produced by a radiologist click is read directly by `gold_to_yolo`. This eliminates the most common source of label-quality degradation: stale annotation exports.

9.2 Why Zero-Initialised FiLM in Practice

The standard alternative to FiLM is concatenation ($\mathbf{P}l' = \text{Conv}\{1 \times 1\}([\mathbf{P}l; \text{tile}])$). Concatenation has two practical problems we explicitly avoid: (a) cold-start instability (a randomly- initialised concat layer perturbs visual features even when text is uninformative), and (b) absence of graceful fallback (the model cannot cleanly fall back to image-only behaviour when the report is empty or wrong). Zero-initialised FiLM resolves both. Crucially, the *safety property* (no degradation at $t = 0$) means a hospital can safely deploy an updated multimodal model knowing that its initial behaviour will not be worse than the image-only model the radiologist team is already familiar with — a prerequisite for non-disruptive clinical adoption.

9.3 Limitations

- **Text granularity.** The text encoder produces a single global vector. Reports describing two distinct lesions ("mass at 2 o'clock and microcalcifications at 8 o'clock") are collapsed. A natural extension is per-FPN-location cross-attention from image tokens to text tokens.
- **Single-view processing.** Each (CC, MLO) view is processed independently. Ipsilateral cross-view attention (cf. our prior BCA- YOLO architecture [author 2026]) is a planned extension.
- **Pseudo-label coarseness.** Stage-2 pseudo-labels are typically $\geq 25\%$ of the breast crop. The three-stage curriculum mitigates this by sequencing pseudo-labels before gold labels — but a Stage-2 model trained without verification is biased toward over-large predictions.
- **Multilingual but not language-aware.** The character-level encoder treats scripts as bucket-hash-of-codepoint. We do not guarantee equal performance across the three scripts, and the required subgroup analysis (§7.4 row 5) is essential.

9.4 Ethical and Deployment Considerations

The system is intended for *triage assist*, not autonomous diagnosis. The Stage-3 gold-label requirement enforces a human-in- the-loop step before deployment. All multilingual text is processed locally; no data leaves the institution; no cloud LLM is called. The pseudo-label generator's `confidence` field, the `needs_radiologist_review` flag, and the FROC operating-point selection (we recommend **Sens@2** FP/image for radiologist- attended workflow) are all designed to support a regulated clinical deployment. The MAMOGRAF platform implements role-based access control (admin / reviewer / annotator), 2FA via

TOTP, JWT auth, rate limiting, audit log of every annotation change with prev/new snapshots, and DICOM de-identification on export.

9.5 Generalisation Beyond Mammography

The closed-loop pattern — preprocess → public bootstrap → text-guided weak labels → radiologist verification UI → gold fine-tune — is not mammography-specific. It generalises to any imaging modality where (i) public bbox-annotated datasets exist, (ii) local clinical reports contain spatial cues, and (iii) local DICOMs have linkable metadata (SOP UID, study UID). Plausible immediate extensions: chest X-ray (public: NIH ChestX-ray14, MIMIC-CXR; spatial cues: laterality, lobe, zone), thyroid ultrasound (public: TNSCUI; spatial cues: laterality, isthmus/upper-pole/mid/lower-pole), and lung CT (public: LIDC-IDRI; spatial cues: lobe, segment).

10. Conclusion

We presented a complete, reproducible, closed-loop system for multilingual mammography lesion detection in low-annotation, multi-script clinical settings. The system integrates four deliverables: an annotation platform (MAMOGRAF), a multilingual text classifier (XS-Classifer), a multimodal one-stage detector (TILLNet-Det) with a novel zero-initialised FiLM fusion operator, and a multilingual weak-supervision pipeline whose output flows directly into a radiologist verification UI that closes the labelling loop. The ten-stage pipeline runner dispatches the entire workflow as one resumable, idempotent command.

The architectural innovation — *zero-initialised FiLM at every pyramid level* — guarantees mathematical equivalence to an image-only baseline at initialisation (Proposition 1, verified to machine precision in §8.4), eliminating cold-start instability and providing a safety property necessary for non-disruptive clinical adoption. The system-level innovation — *embedding the verification UI inside the production annotation platform* — eliminates the most common source of label-quality degradation in medical-AI deployments: stale ETL between research and production tooling.

To our knowledge, this is the first published mammography CAD system that simultaneously (a) handles three co-existing scripts in a unified character-level encoder, (b) closes the labelling loop through an integrated verification UI, (c) provides a mathematically non-degrading multimodal fusion operator, and (d) ships a fully reproducible end-to-end pipeline tailored to a low-annotation, multi-script reality. The implementation is released open-source.

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[To be added.]

Conflicts of Interest

[To be added.]

Data and Code Availability

CBIS-DDSM is available from The Cancer Imaging Archive (<https://www.cancerimagingarchive.net/collection/cbis-ddsm/>). Local clinical data cannot be shared due to patient confidentiality. The full implementation — preprocessing, CBIS-DDSM converter, multilingual weak-supervision pipeline, MAMOGRAF platform, TILLNet-Det model, trainer, FROC evaluator, gold-label converter, ten-stage pipeline runner — is released open-source under [licence] at [URL].

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